MIL-S-20660C
15 February 1977
SUPERSEDING
MIL-S-20660B
19 September 1967

MILITARY SPECIFICATION

STEREOSCOPE, LENS-PRISM-MIRROR, AERIAL PHOTOGRAPH INTERPRETATION (WITH BINOCULARS AND CASE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mirror stereoscope with binoculars and a carrying case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 20060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-A-225	- Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for.
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-M-13508	- Mirror, Front Surfaced Aluminized: for Optical Elements.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-150	- Photographic Lenses.

(Copies of specifications and standards required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The stereoscope shall consist of lens, prisms, mirrors, binoculars, and folding legs (see 6.3 and Figure 1). The stereoscope shall accommodate photographs up to 9 by 9 inches.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Aluminum castings. Aluminum castings shall be sand castings conforming to QQ-A-601, Alloy 356.0, Temper T6, or die castings conforming to QQ-A-591, Alloy No. 413.0 or 360.0, or 380.

3.3 Frame. The frame of the stereoscope shall be a single aluminum casting of such strength and rigidity as to firmly support the optical parts and detachable binoculars, and to serve as a headrest for the operator.

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3.3.1 Legs. Four folding legs shall be attached to the frame of the stereoscope. The legs shall be of sufficient strength to rigidly support the frame while the stereoscope is in use. One leg shall be adjustable in length so that the stereoscope will be firmly supported on slightly irregular flat surfaces. The legs shall be aluminum conforming to QQ-A-225, Temper T4, Alloy 6061; QQ-A-601, Alloy 356.0, Temper T6; or, QQ-A-591, Alloy No. 413.0 or 360.0, or 380.

3.3.2 Handles. A corrosion-resisting handle shall be provided on each end of the frame to facilitate lifting and moving the stereoscope.

3.4 Optical parts. Plaster or other forms of setting cement shall not be used for mounting the lenses, prisms, or the mirrors.

3.4.1 Lenses. The lenses, exclusive of the binoculars, shall have focal lengths of 11.25 inches, plus or minus 1 percent. The lenses shall be free of striae, bubbles, scratches, or other defects that adversely affect optical quality. The image presented through the lenses shall not be perceptibly affected by spherical or chromatic aberration, coma, astigmatism, or distortion. The right and left lenses shall be interchangeable. When mounted, the lenses shall be of sufficient size to permit easy viewing by operators having interpupillary distances from 52 to 75 millimeters (mm).

3.4.1.1 Lens mounting. The lenses shall be mounted in rings so that they will be firmly held in place, but in a manner that will permit easy removal for replacement. The lenses shall be sealed to prevent ingress of dirt, dust, or other foreign material between the lenses and the prisms. The axes of the lenses shall be spaced 66 mm apart, plus or minus 2 mm. The optical distance from the lenses to the plane of the photographs on both sides shall be equal to each other within 1 percent, and the distance shall be such that the eyes of an observer are accommodated to a distance between 8 feet and infinity.

3.4.2 Mirrors. The mirrors shall conform to MIL-M-13508. The mirrors shall be 0.125 inch thick and all edges shall be ground. Easily attachable covers of soft material, which is fungus resistant or treated to be fungus resistant, shall be provided to protect the mirrors when not in use.

3.4.2.1 Mirror mounting. The mirrors shall be mounted on the frame by adjustable spring clamps at the corners. The mounting shall not cause strain on the mirrors and shall permit easy removal or replacement of mirrors. The right and left mirrors shall be interchangeable.

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3.4.3 Prisms. The prisms shall be high-grade, commercial, right-angle, 45-degree prisms, free from color, and with no visible striae, streaks, cords, or other defects that would impair the vision in normal use. The ends shall be fine ground with the edges beveled.

3.4.3.1 Prism mounting. The prism shall be firmly held in place by means of cadmium-plated steel springs and shall be removable and interchangeable. The prisms shall in no way be exposed except insofar as is optically necessary.

3.5 Optical alinement. The right and left optical systems, exclusive of the binoculars, shall have their axes parallel within one degree of divergence towards the observer. This alinement shall be accomplished and sealed at the time of manufacture.

3.6 Reflection-reducing coating. All air-glass surfaces of the lenses and the prisms used in the stereoscope and the binoculars shall be coated with a reflection-reducing film conforming to MIL-C-675.

3.7 Magnifying system. The magnifying system shall be binocular; each ocular shall consist of a 20-mm objective lens, two erecting prisms, and an eye lens such that, in combination with the optics of the basic stereoscope, a magnification of about four power is obtained. Exit pupil size of each ocular shall be 5 mm, and each field of view shall be not less than 1.625 inches in diameter at the plane of the photographs. The image presented to an observer shall be erect. The binoculars shall be constructed so as to accommodate interpupillary distances and shall have sufficient range of movement to give plus or minus five diopters correction with a scale showing departure in diopters from infinite accommodation.

3.7.1 Binocular mounting. The binoculars shall be mounted so that they may swing into or out of position for viewing without being detached, and so that photographs properly oriented for viewing with binoculars need not be reoriented for viewing without them. When the binoculars are attached but not in use, their supporting plate shall provide a convenient headrest for viewing without magnification. The binoculars shall be easily detachable without necessitating the use of tools or loosening of thumbscrews or wing nuts, and shall not come loose under normal usage. The oculars of the magnifying system shall have their axes parallel within 8 minutes of convergence toward the observer for all interpupillary settings and, when in position for viewing, their axes shall be coincident with the lens axes of the basic stereoscope.

3.7.2 Interchangeability. The binocular magnifiers of all stereoscopes of this type fabricated by the same manufacturer shall be interchangeable.

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3.8 Resolving power. The resolving power of the stereoscope lenses, without the binoculars, shall be at least 6 lines per millimeter at the center of the field of view. The resolving power of the binoculars in combination with lenses of the basic stereoscope shall be not less than 14 lines per millimeter.

3.9 Case. Each stereoscope shall be provided with a fitted carrying case that will contain the complete stereoscope. The weight of the case without the instrument shall be not more than 8 pounds. The case shall be provided with a carrying handle. Two positive-acting fasteners shall be provided for keeping the lid tightly closed. If the case is made of wood, the inside shall be given two coats of spar varnish. Compartments inside the case shall be lined with a material which is fungus resistant or treated to be fungus resistant and thick enough to protect the instrument from damage during transportation. The outside of the case shall be finished at the contractor's option.

3.10 Identification marking. Each stereoscope and its binocular shall be identified in accordance with MIL-STD-130. The marking shall be applied to the stereoscope on plates conforming to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material.

3.11 Treatment and painting. All parts of the equipment normally painted shall be cleaned, treated, and painted. Treatment and painting shall be at the contractor's option, except that surfaces likely to reflect detrimental light into the optical system shall be painted a dull nonreflecting black.

3.12 Workmanship. All parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 and 4.4.2.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The stereoscopes shall be examined as specified herein for the following defects:

Major

101. Stereoscope does not accommodate 9- by 9-inch photographs.
102. Materials not as specified.
103. Optical parts cemented or plastered in place.
104. Lenses, prisms, and mirrors not interchangeable or not mounted as specified.
105. Reflection-reducing coating not as specified.
106. Binoculars do not swing freely into and away from the viewing position.
107. Binoculars are not easily detachable.
108. Binocular magnifiers not interchangeable.
109. Stereoscope carrying case not as specified.

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Minor

201. Legs do not fold freely and easily.
202. One leg not adjustable.
203. Mirror protectors missing.
204. Carrying case not padded to protect the stereoscope from damage.
205. Identification marking missing, incomplete, or illegible.
206. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Resolving power without binoculars. The stereoscope shall be set up in viewing position with the binoculars folded out of viewing position, or the binoculars may be detached from the stereoscope. A standard resolving power test high contrast target in accordance with MIL-STD-150 shall be placed in viewing position beneath the stereoscope, first on one side, then on the other. An observer with normal 20/20 or corrected 20/20 vision shall view the target first through one side of the stereoscope and then through the other side to determine conformance to 3.8. The target shall be viewed near the center of the field of view and throughout the usable area. Failure to clearly resolve at least 6 lines per millimeter shall constitute failure of this test.

4.4.2.2 Resolving power with binoculars. The stereoscope and target shall be set up in viewing position according to 4.4.2.1 and the binoculars shall be placed in viewing position. Each ocular shall be focused individually on the target. The target shall be viewed through one ocular at a time to determine conformance to 3.8. Failure to clearly resolve at least 14 lines per millimeter shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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110. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
111. Box closure not in accordance with Method V as specified.
112. Blocking, bracing, and cushioning not provided as specified.
113. Strapping not as specified.
114. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each stereoscope shall be preserved in accordance with MIL-P-116, Method IIb. The carrying case shall be the inner container, and the outer container shall be a box conforming to PPP-B-636, W5c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each stereoscope shall be preserved and packaged in a manner to assure protection against deterioration and damage from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Stereoscopes, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Blocking, bracing, and cushioning shall be provided to prevent movement of items within the box. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Stereoscopes, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be legible and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The stereoscope is used for the examination of stereoscopic pairs of aerial photographs. The binoculars are used with the stereoscope to magnify portions of the aerial photograph for closer examination.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(c) When other than Finish B strapping is required (see 5.2.1).

6.3 Information figure. Figure 1 shows a type of stereoscope which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other stereoscopes which conform to this specification.

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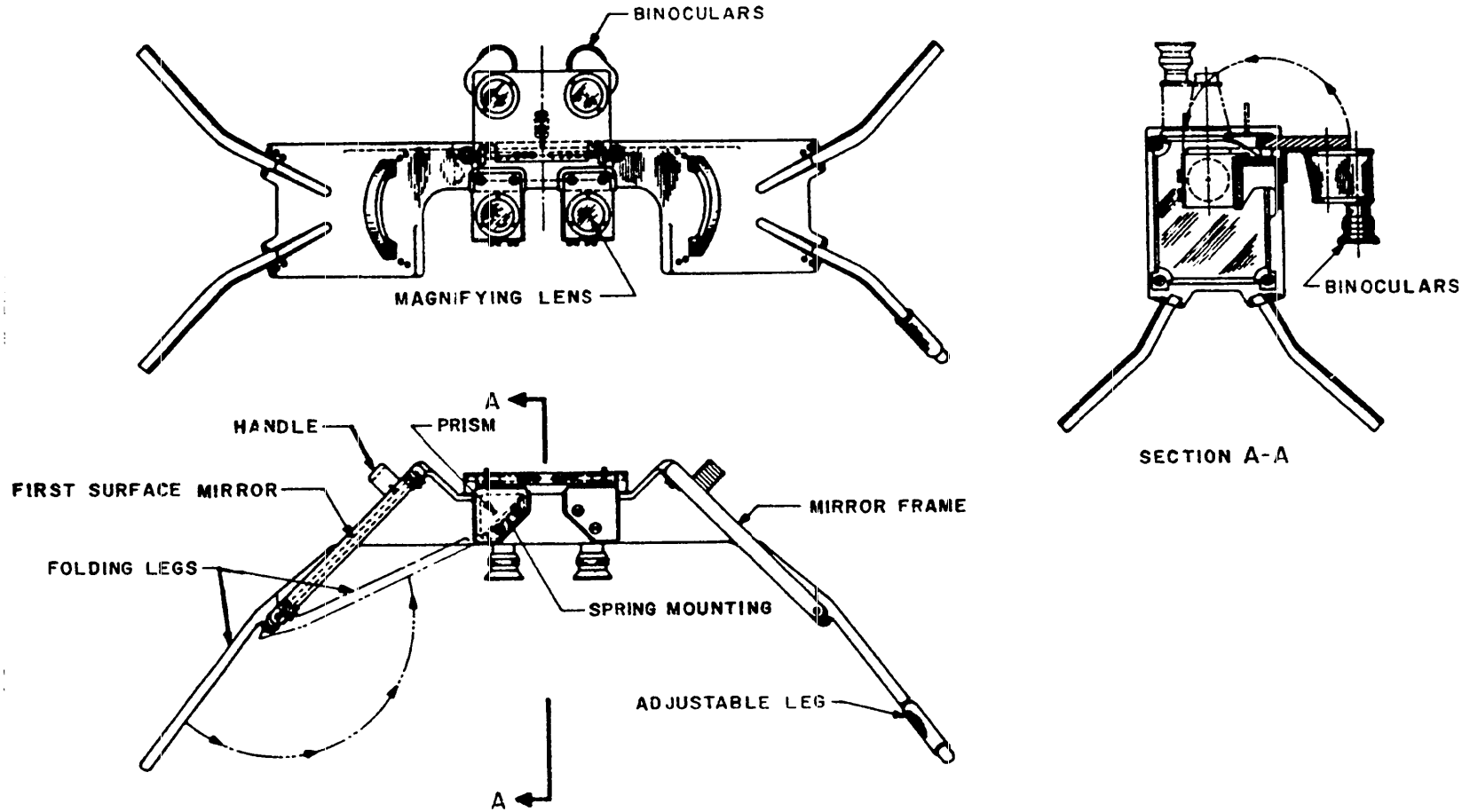


FIGURE 1. Stereoscope general assembly.

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DOCUMENT IDENTIFIER AND TITLE MIL-S-20660C Stereoscope, Lens-Prism-Mirror, Aerial Photograph Interpretation (with Binoculars and Case)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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